



Fortinet Cybersecurity Academy: Configuring a FortiGate 40F Firewall for a SOHO Environment

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Purpose

This lab is intended to introduce the Fortinet ecosystem, providing basic knowledge of the FortiGate firewall console and GUI interface. The lab also provides insight into the default configuration of a FortiGate firewall and how it can be used for SOHO environment.

Background

SOHO, short for Small Office/Home Office, is a network type commonly used by individuals or small businesses with less than 10 employees. This network type commonly uses smaller-scale routers, switches, and firewalls compared to their large enterprise counterparts. SOHO networks provide numerous advantages to teams of 1-10 people as they are easier to set up and are more affordable than full-size network equipment. SOHO networks often only have a single router, and may contain switches, wireless access points, and end devices such as computers and printers.

Fortinet is a cybersecurity company founded in 2000 in Sunnyvale, CA. They are known for their flagship product, the FortiGate firewall, as well as a wide variety of other networking and security devices, such as the FortiSwitch and the FortiAP, and services such as FortiSandbox, FortiAuthenticator, FortiVoice, and FortiDDoS. Fortigate is an S&P 500 component and is listed on the NASDAQ as \$FTNT.

The FortiGate 40-F is a firewall developed by Fortinet. It has capabilities expected of a modern firewall such as full routing capability, DHCP server capability, and support for a variety of filtering methods. The 40-F also supports running its own local RADIUS server with a feature called Local Auth (Authentication). The 40-F uses a fanless design, allowing it to operate silently. The 40-F has a small form factor at 1.5 x 8.5 x 6.3 inches, meaning it can easily fit into existing networking setups. By default, the 40-F gives out DHCP addresses in the 192.168.1.0/24 subnet to its clients (from .110-.210, specifically) and its GUI client can be accessed via HTTPS at 192.168.1.99.

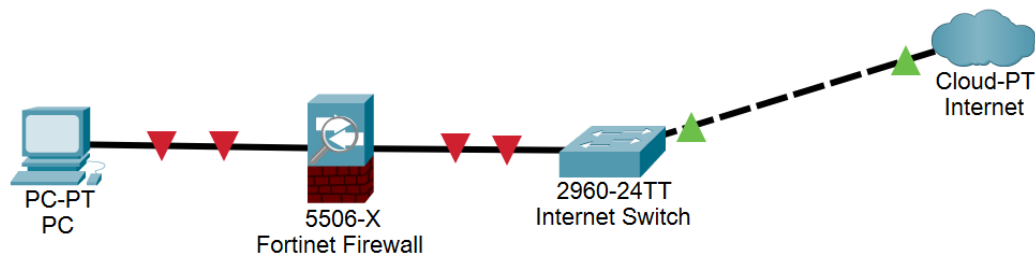
The FortiAP 421E is an access point developed by Fortinet. It has 8 internal antennae and 2 internal radios broadcasting on 2.4 GHz and 5 GHz respectively. It supports a variety of authentication methods, including WPA and WPA2 with either 802.1X or PSK, WEP, and a MAC blacklist/whitelist. It can form a mesh with other FortiAP devices to ensure more uniform connectivity across a site. The 421E can also handle a maximum of 512 clients per radio split across a maximum of 16 SSIDs.

Comparing Fortinet to other firewall vendors, Fortinet heavily focuses on integrating products together. Furthermore, using multiple Fortinet products will further increase security and threat

detection. It isn't only suitable for SOHO environments, but also enterprise-grade solutions for large organizations.

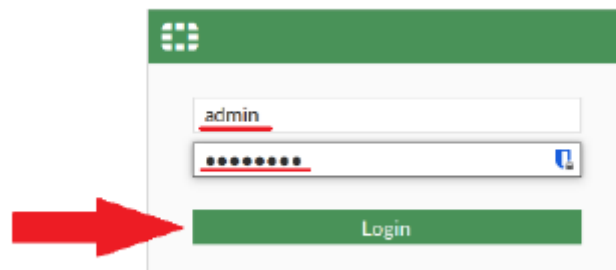
Fortinet also directly competes with the leading competitors like Cisco, Palo Alto Networks, and Check Point. Combining many different Fortinet systems and ecosystems will grant you access to many different features.

Topology

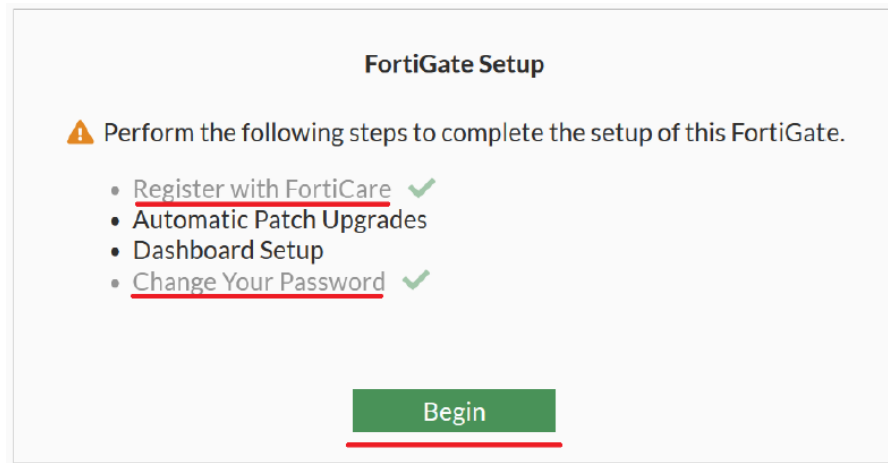


Lab Commands

Connect your PC to the firewall via ethernet. Go to <https://192.168.1.99> and log in with the username and password you set.

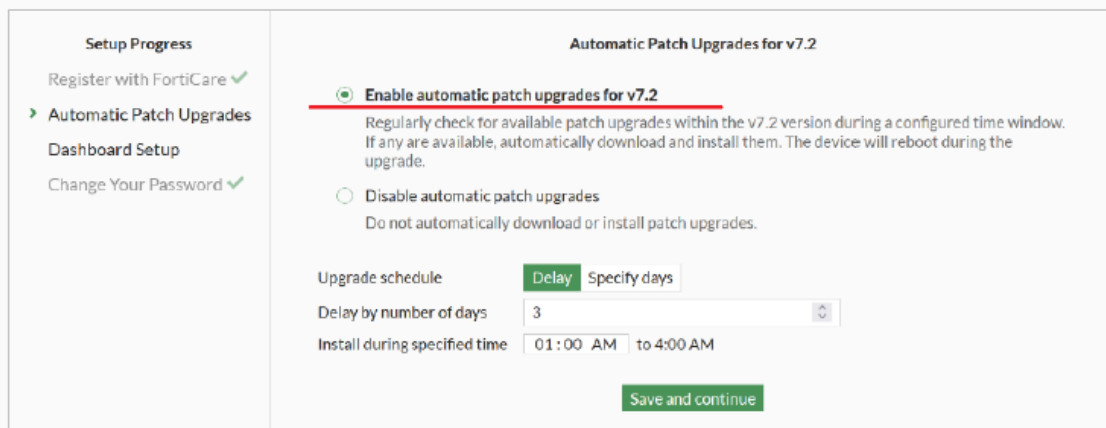


You should see the following setup screen. Note that the Change Your Password checklist item has already been completed, as we completed this earlier. Click Begin.



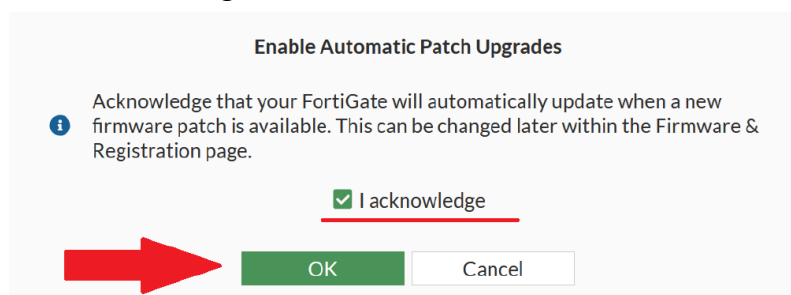
The screenshot shows the 'FortiGate Setup' window. At the top, it says 'FortiGate Setup'. Below that, an orange warning icon is followed by the text 'Perform the following steps to complete the setup of this FortiGate.' A checklist follows with four items: 'Register with FortiCare' (with a green checkmark), 'Automatic Patch Upgrades', 'Dashboard Setup', and 'Change Your Password' (with a green checkmark). The 'Automatic Patch Upgrades' item is highlighted with a red underline. At the bottom center, there is a green 'Begin' button with a red underline.

Click Enable Automatic Patch Upgrades and leave the other settings as default.



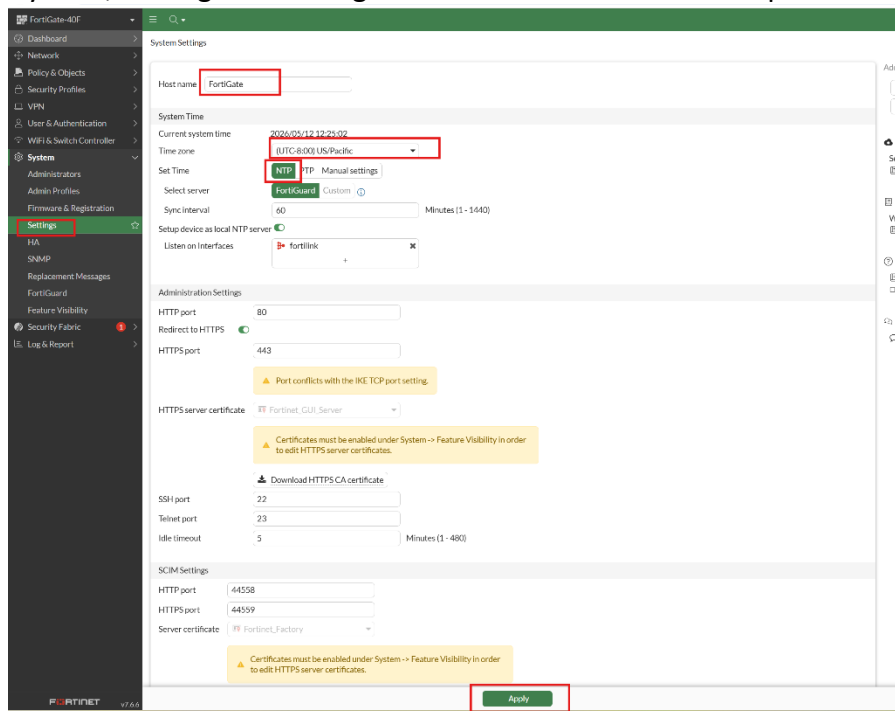
The screenshot shows the 'Automatic Patch Upgrades for v7.2' configuration screen. On the left, a 'Setup Progress' sidebar lists 'Register with FortiCare' (checked), 'Automatic Patch Upgrades' (selected with a green arrow), 'Dashboard Setup', and 'Change Your Password' (checked). The main area has the title 'Automatic Patch Upgrades for v7.2'. It features two radio buttons: 'Enable automatic patch upgrades for v7.2' (selected and underlined in red) and 'Disable automatic patch upgrades'. Below the radio buttons, there is an 'Upgrade schedule' section with a 'Delay' button selected and a 'Specify days' button. Under 'Delay', there is a 'Delay by number of days' field set to '3' and an 'Install during specified time' field set to '01:00 AM to 4:00 AM'. At the bottom right, there is a green 'Save and continue' button.

Click I acknowledge and click OK.

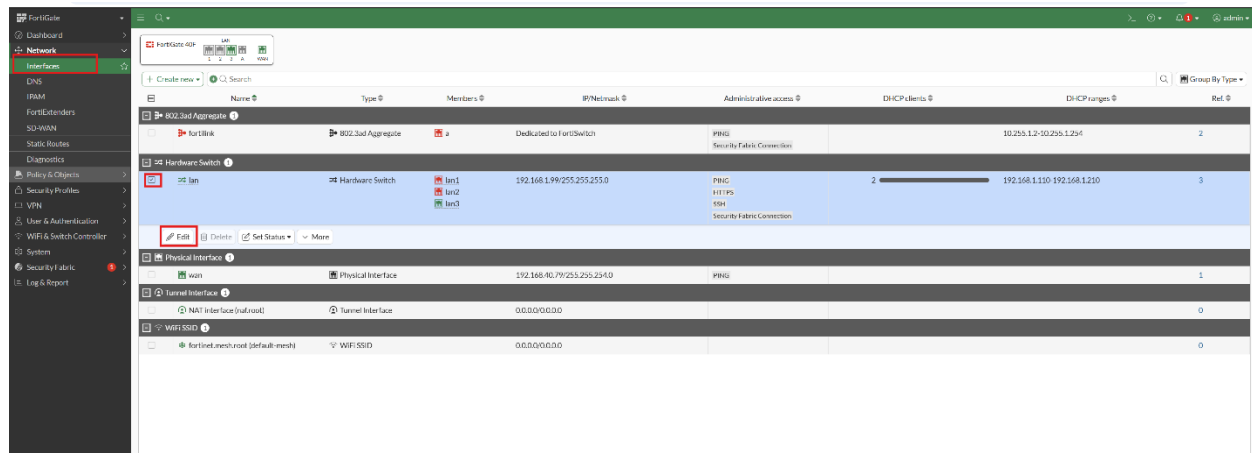


The screenshot shows a dialog box titled 'Enable Automatic Patch Upgrades'. It contains an information icon and the text: 'Acknowledge that your FortiGate will automatically update when a new firmware patch is available. This can be changed later within the Firmware & Registration page.' Below this text, there is a checkbox labeled 'I acknowledge' which is checked and underlined in red. At the bottom, there are two buttons: 'OK' and 'Cancel'. A large red arrow points to the 'OK' button.

To change the hostname and configure the system time of the Fortinet firewall, head over to System, Settings and configure the Host Name and the respective time zones.



Next, head over to Network > Interfaces. Click on edit to configure the different parts of the network. First we will do the LAN interface.



Here you can configure the LAN configuration settings. For this lab, we will be configuring its Alias to Office LAN. You can also optionally change the IP address of the web-based firewall management interface to an IP address of your choice by configuring the IP/Netmask into your desired IP address. For this lab, we changed from the default IP address of 192.168.1.99/24 to 192.168.10.1/24. For DHCP, you can configure the address range for leasing. For demonstration, we have configured the address range of leasing from 192.168.10.10 to 192.168.10.200. Don't forget to set default gateway to the configured IP/Netmask. When you are finished, press OK to finalize your changes.

Edit Interface

Name:

Alias:

Type:

Interface members:

Role:

Address

Addressing mode: ☒ Manual ☐ PAM ☐ DHCP ☐ PPPoE

IP/Netmask:

Create address object matching subnet: ☒

Name:

Destination:

Secondary IP address: ☐

Administrative Access

IPv4: ☒ HTTPS ☐ HTTP ☒ PING ☐ SSH ☐ SNMP ☒ Security Fabric ☐ FMG-Access ☐ RADIUS Accounting ☐ Connection ☐ FTM ☐ SCIM ☐ Speed Test

Receive LLDP: ☒ Use VDOM Setting ☐ Enable ☐ Disable

Transmit LLDP: ☒ Use VDOM Setting ☐ Enable ☐ Disable

DHCP Server

DHCP status: ☒ Enabled ☐ Disabled

Address range:

Netmask:

Default gateway: ☒ Specify

DNS server: ☒ Same as System DNS ☐ Same as Interface IP

Lease time: ☒ 604800

FortiClient On-Net Status: ☒ Default

Advanced

FortiGate

FortiGate

Active Administrator Sessions

HTTPS

Status

Up

MAC address

84:39:8f:5:3a:29

Additional Information

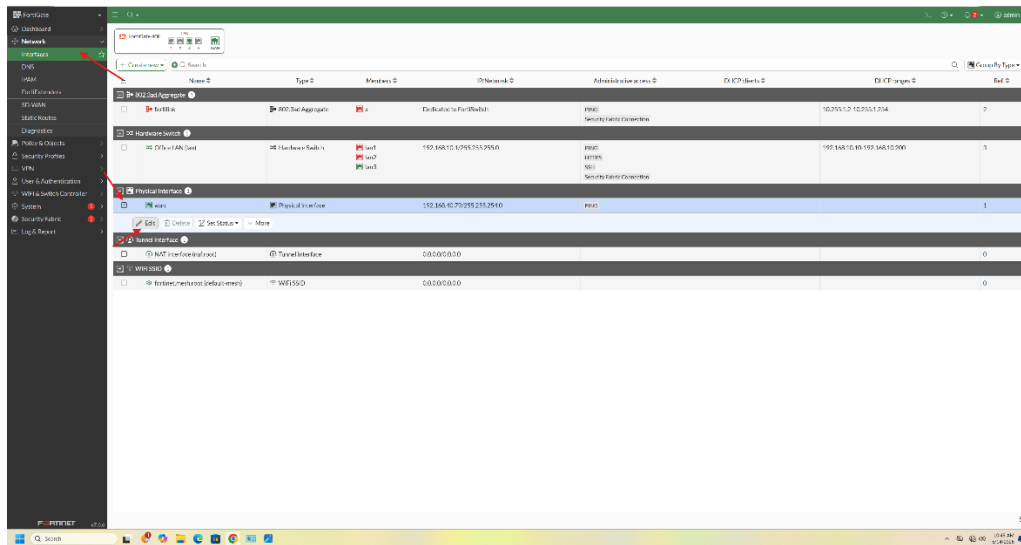
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Similarly, by selecting the WAN and clicking on edit, we can also configure its Alias and interface members.



Name	wan	
Alias	Internet	
Type	Physical Interface	
Role ⓘ	WAN	
Estimated bandwidth ⓘ	0	kbps Upstream
	0	kbps Downstream

Press ok to save

Administrative Access

IPv4

☐ HTTPS ☐ HTTP ⓘ ☒ PING

☒ FMG-Access ☐ SSH ☐ SNMP

☐ FTM ☐ RADIUS Accounting ☐ Security Fabric Connection ⓘ

☐ Speed Test

Receive LLDP ⓘ Use VDOM Setting Enable Disable

Traffic Shaping

Outbound shaping profile ☐

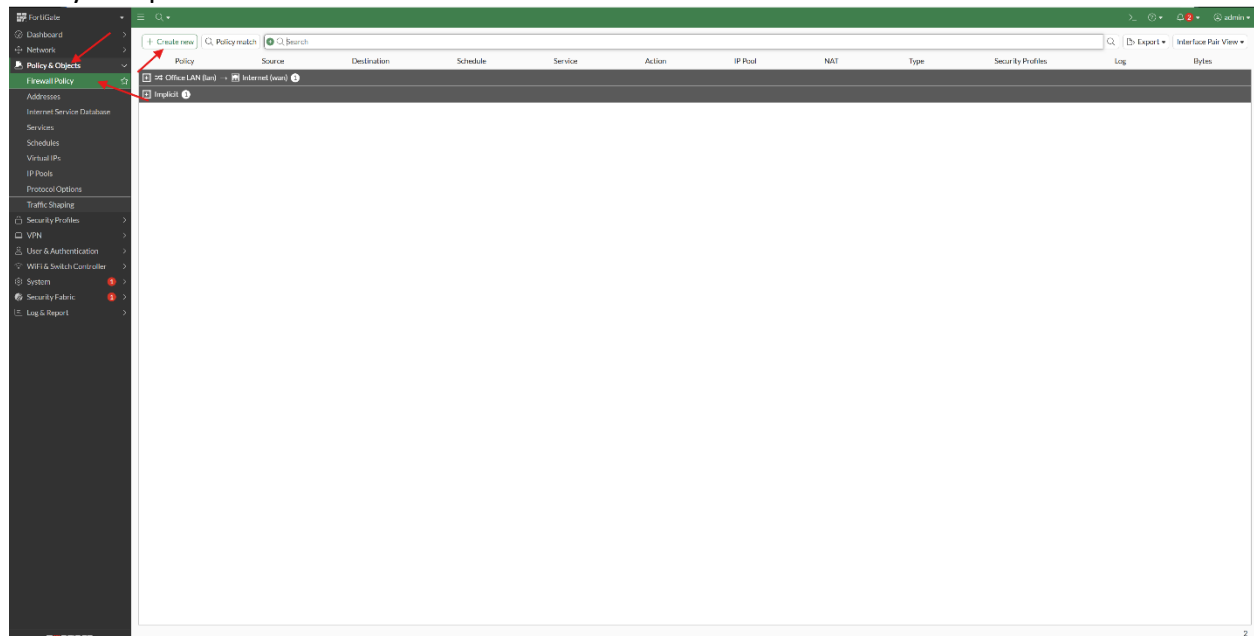
Miscellaneous

Comments 0/255

Status Enabled Disabled

OK Cancel

To configure your LAN-to-WAN firewall policy, head over to Policy & Objects, Firewall Policy and press Create New.



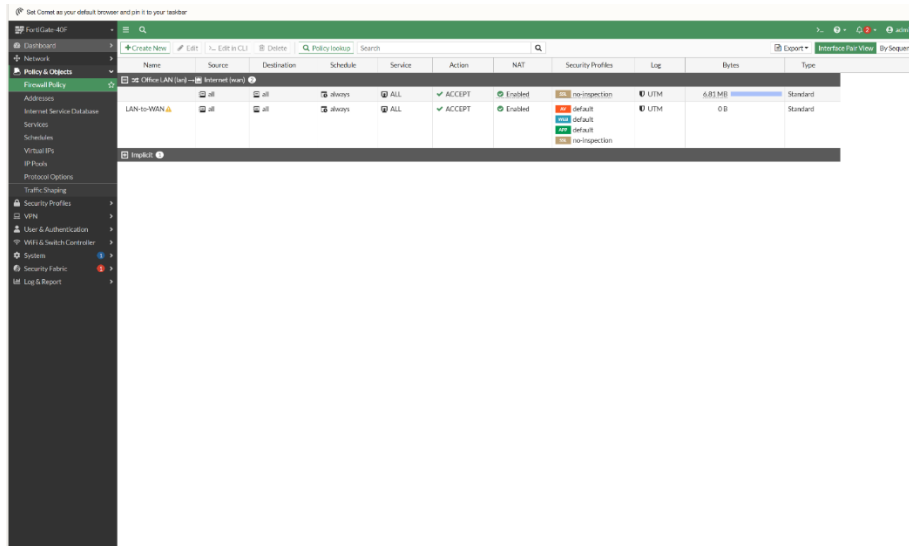
Here, configure these configurations. This will allow LAN to WAN policies that direct traffic from all sources to all destinations, with NAT configurations. When you are done, press OK.

The screenshot shows the 'Create New Policy' configuration window in Fortinet's management interface. Red arrows highlight the following settings:

- Name:** LAN to WAN
- Schedule:** always
- Action:** ACCEPT (checked), DENY (unchecked)
- Incoming interface:** Office LAN (lan)
- Outgoing interface:** Internet (wan)
- Source:** all
- User/group:** (empty)
- Destination:** all
- Service:** ALL
- NAT:** Use Outgoing Interface Address (checked), Use Dynamic IP Pool (unchecked)
- Source port translation:** Always (checked), When port conflicts (unchecked), Never (unchecked)
- Protocol options:** pfs: default
- Security Profiles:**
 - AntiVirus: av: default
 - Web filter: web: default
 - DNS filter: (empty)
 - Application control: app: default
 - IPS: (empty)
 - SSL inspection: ssl: no-inspection
- Logging Options:** Log allowed traffic: Security events (checked), All sessions (unchecked)
- Comments:** (empty)
- Enable this policy:** (checked)

The 'OK' button is highlighted with a red arrow at the bottom of the window.

When finished, your screen should look like this.



Your done with your soho/

Problems

We found that the FortiAP cannot be plugged directly into the FortiGate without a separate power source, as the FortiGate does not provide power over ethernet. We fixed this by placing a standard 48V PoE injector in between the firewall and the access point.

One other problem that we had was that after we manually set the Ips to our ethernet and our default gateway to the IP address of the Fortinet, after we set up the SOHO configuration, it didn't work. We looked at Ipconfig /all and realized that we aren't getting DNS addresses. It was because we are supposed to be on DHCP automatic addressing. After we switched to DHCP, it worked.

Conclusion

To wrap up, I now have a strong understanding of the basic Fortinet ecosystem, the FortiOS GUI interface, and various WPA2 security policies. I am now confident that I could replicate this setup in a real SOHO environment.